

Sampling Variability and the Central Limit Theorem

Why samples differ, and why averages become predictable

Sergio I. García-Ríos

PA 311N: Data Analysis for Public Policy

LBJ School of Public Affairs

October 5, 2026

Where we are going

Last week we asked how probability models describe **individual observations**.

Today we ask a different question:

If we repeatedly draw samples,
how will our estimates vary?

That question is the bridge from descriptive statistics to statistical inference.

The puzzle behind statistical inference

Public policy rarely gives us the whole population.

- ▶ We want a **population parameter**: μ , p , a difference, a regression coefficient.
- ▶ We observe a **sample statistic**: $\bar{x}$, $\hat{p}$, and so on.
- ▶ A different random sample would usually give a different statistic.

The puzzle

If estimates change from sample to sample, why should we trust any one sample?

Today: because the variation itself has a pattern we can model.

Population parameter vs. sample statistic

Population

All 2,930 residential home sales in the Ames data.

$$\mu = 1499.69 \text{ sq. ft.}$$

$$\sigma = 505.51 \text{ sq. ft.}$$

One random sample

Suppose we observe only $n = 50$ homes.

$$\bar{x} = \text{mean area in those 50 homes}$$

The value of $\bar{x}$ depends on which homes were sampled.

μ is fixed. $\bar{x}$ varies.

One population, many possible samples

Suppose two analysts independently draw 50 homes from the same population.

Analyst A

A random sample might produce

$$\bar{x} = 1417 \text{ sq. ft.}$$

Analyst B

Another random sample might produce

$$\bar{x} = 1660 \text{ sq. ft.}$$

Both analysts used the same population and the same sample size.

Different estimates are an expected consequence of random sampling.

Sampling variability is not automatically an error

Sampling variability

Natural variation in a statistic because different random samples contain different observations.

Bias or bad measurement

A systematic problem that pushes estimates in a particular direction.

A perfectly designed random sample still has sampling variability.

Your Turn

If two high-quality polls of the same population report 47% and 49%, must one of them be wrong?

**We need a distribution
of the statistic itself.**

Three different distributions

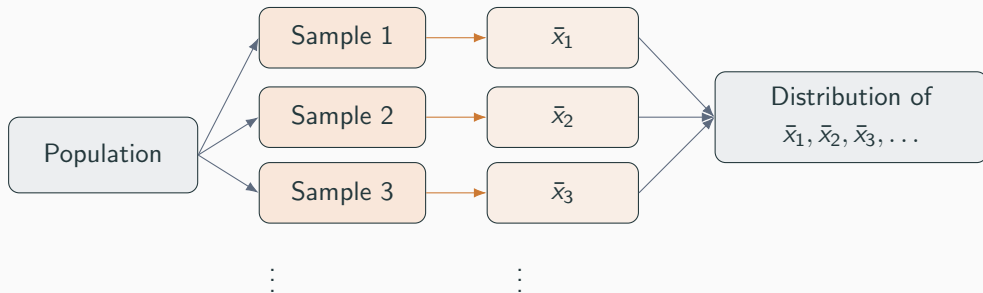
1. **Population distribution:** values for every unit in the population.
2. **Sample distribution:** observed values in one particular sample.
3. **Sampling distribution:** values of a statistic across many possible samples of the same size.

For the mean:

sampling distribution of $\bar{x}$

The observations in a sampling distribution are **sample means**, not individual homes or people.

How a sampling distribution is constructed



Same population. Same n . New random sample each time.

Your turn: what does each dot represent?

Your Turn

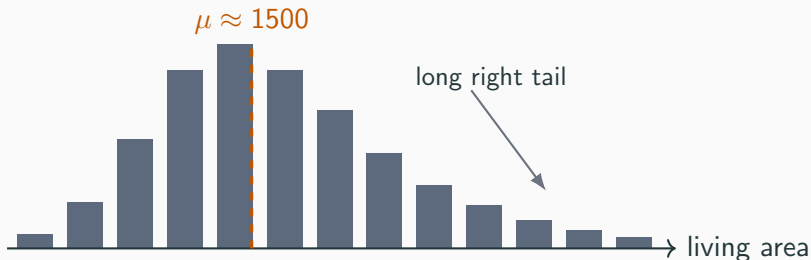
Imagine a histogram built from 5,000 values of $\bar{x}$, each calculated from a random sample of 50 Ames homes.

What does **one observation** in that histogram represent?

One **sample mean** from one random sample of 50 homes.

Ames gives us a useful experiment

The population distribution of living area is **right-skewed**:



$$\mu = 1499.69, \quad \sigma = 505.51.$$

This is useful because the raw population is *not* perfectly normal.

Standard deviation vs. standard error

Standard deviation, σ

Describes variation among **individual observations** in the population.

For Ames area:

$$\sigma \approx 505.5 \text{ sq. ft.}$$

Standard error, $SE(\bar{x})$

Describes variation among **sample means** across repeated samples.

For $n = 100$:

$$SE(\bar{x}) \approx 50.6 \text{ sq. ft.}$$

SD describes people/homes/units. SE describes estimates.

The standard error of the sample mean

When observations are independent,

$$\text{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

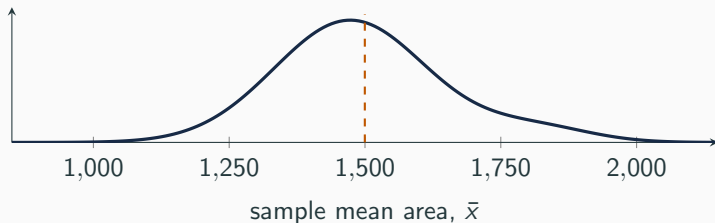
- ▶ More population variability (σ larger) $\Rightarrow$ more uncertainty.
- ▶ Larger sample size (n larger) $\Rightarrow$ less uncertainty.
- ▶ The benefit of sample size follows a **square-root law**.

In practice σ is usually unknown; later we will estimate it using s .

Sampling distribution for $n = 10$

For samples of 10 homes:

$$SE(\bar{x}) = \frac{505.51}{\sqrt{10}} \approx 160 \text{ sq. ft.}$$

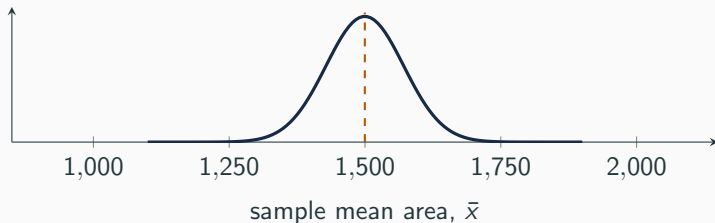


The distribution is centered near the population mean, but with $n = 10$ some skew may remain.

Sampling distribution for $n = 50$

For samples of 50 homes:

$$SE(\bar{x}) = \frac{505.51}{\sqrt{50}} \approx 71.5 \text{ sq. ft.}$$

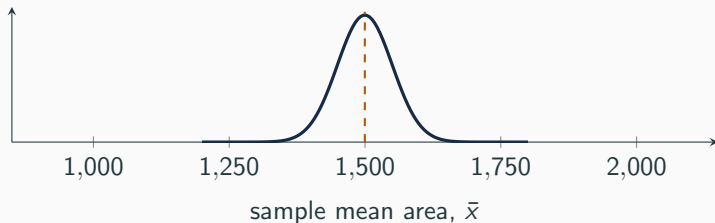


Same center. Much less variability among sample means.

Sampling distribution for $n = 100$

For samples of 100 homes:

$$SE(\bar{x}) = \frac{505.51}{\sqrt{100}} \approx 50.6 \text{ sq. ft.}$$



Larger samples produce more stable estimates.

What changes as n increases?

	$n = 10$	$n = 50$	$n = 100$
Center	≈ 1500	≈ 1500	≈ 1500
SE	≈ 160	≈ 71.5	≈ 50.6
Shape	some skew may remain	closer to normal	near normal

Three recurring features

The sampling distribution has a predictable **center**, **spread**, and, under the right conditions, **shape**.

**The spread of a sampling distribution
has its own name: standard error.**

The square-root law

To cut the standard error in half, how much larger must the sample be?

$$SE = \frac{\sigma}{\sqrt{n}}$$

If we replace n with $4n$:

$$SE = \frac{\sigma}{\sqrt{4n}} = \frac{\sigma}{2\sqrt{n}}.$$

4 times the sample size $\Rightarrow$ half the SE

Precision improves with sample size, but with diminishing returns.

Your turn: compare precision

Your Turn

A policy analyst estimates mean processing time. Assume the population SD is 24 days.

Compare the standard error when:

$$n = 36 \quad \text{versus} \quad n = 144.$$

$$SE_{36} = \frac{24}{6} = 4 \text{ days}$$

$$SE_{144} = \frac{24}{12} = 2 \text{ days}$$

Quadrupling n cuts the standard error in half.

**The Central Limit Theorem
explains the shape.**

Central Limit Theorem: the headline

Under appropriate conditions, the sampling distribution of the sample mean is approximately normal:

$$\bar{X} \approx N\left(\text{mean} = \mu, \text{SD} = \frac{\sigma}{\sqrt{n}}\right)$$

So the CLT tells us three things:

- ▶ **Center:** $E(\bar{X}) = \mu$
- ▶ **Spread:** $\text{SE}(\bar{X}) = \sigma/\sqrt{n}$
- ▶ **Shape:** approximately normal when conditions are adequate

What the CLT does not say

Common misconception

The CLT does **not** say that the original population becomes normal as n increases.

It says the distribution of **sample means** becomes approximately normal.



When can we use the CLT?

Two ideas matter most.

1. Independence

- Random sampling helps.
- If sampling without replacement, a common check is $n < 10\%$ of the population.

2. Population shape and sample size

- Nearly normal population: small samples can work well.
- Moderate skew: larger n improves the approximation.
- Strong skew or extreme outliers: we may need a much larger n .

There is no universal magic cutoff where the CLT suddenly turns on.

Why “ $n = 30$ ” is only a rule of thumb

Mildly skewed population

A sample size around 30 may already give a useful normal approximation for $\bar{x}$.

Very skewed population

A sample size of 30 may still be too small if rare extreme values dominate the distribution.

The question is not simply “Is $n \geq 30$?”

It is: Is the approximation plausible for this data-generating process?

The key contrast from Lab 4

Last Wednesday we asked about an **individual** Ames home:

$$P(X \geq 2000).$$

Because home area is right-skewed, the normal model for individual homes is imperfect.

Today we ask about the **mean of 100 homes**:

$$P(\bar{X} < 1450).$$

The CLT can make a normal approximation useful even though the original population is skewed.

Individual observations and sample means have different distributions.

Our turn: probability for a sample mean

For Ames home area,

$$\mu = 1499.69, \quad \sigma = 505.51, \quad n = 100.$$

By the CLT,

$$\begin{aligned}\bar{X} &\approx N\left(\text{mean} = 1499.69, \text{SD} = \frac{505.51}{\sqrt{100}}\right) \\ &= N(\text{mean} = 1499.69, \text{SD} = 50.55).\end{aligned}$$

What is $P(\bar{X} < 1450)$?

$$z = \frac{1450 - 1499.69}{50.55} \approx -0.98 \quad \Rightarrow \quad \boxed{P(\bar{X} < 1450) \approx 0.163}$$

What does 0.163 mean?

Our Turn

Imagine repeatedly taking random samples of 100 Ames homes and calculating the mean living area each time.

A probability of about 0.163 means:

About 16% of those samples would have a sample mean below 1,450 square feet.

It does **not** mean 16% of individual homes are below 1,450 square feet.

The same calculation in R

Once the sampling distribution is identified, R can calculate the tail area directly.

```
mu <- 1499.69
sigma <- 505.51
n <- 100

se <- sigma / sqrt(n)

pnorm(1450, mean = mu, sd = se)
# about 0.163
```

The difficult part is usually not typing `pnorm()`. It is deciding **which distribution** and **which standard deviation** belong in the calculation.

Transfer the idea to a policy setting

Suppose a public benefits agency has processing times with

$$\mu = 42 \text{ days}, \quad \sigma = 18 \text{ days}.$$

A manager reviews a random sample of $n = 81$ cases, so

$$\text{SE}(\bar{X}) = \frac{18}{\sqrt{81}} = 2 \text{ days}.$$

What is the probability the sample average exceeds 46 days?

$$z = \frac{46 - 42}{2} = 2 \quad \implies \quad P(\bar{X} > 46) \approx 0.023$$

Interpretation

A sample mean above 46 days would be relatively unusual if the long-run mean were really 42 days.

Your turn: change the sample size

Your Turn

Keep $\mu = 42$ and $\sigma = 18$, but now sample only $n = 36$ cases.

1. What is the standard error?
2. Is $P(\bar{X} > 46)$ larger or smaller than it was for $n = 81$?
3. Why?

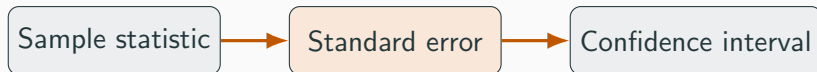
$$SE = \frac{18}{6} = 3, \quad z = \frac{46 - 42}{3} = 1.33, \quad P(\bar{X} > 46) \approx 0.092.$$

Why did the probability grow?

Smaller samples produce more variable means, so an average above 46 days is less unusual.

Why this matters for inference

A point estimate alone does not tell us how precise it is.



A rough 95% pattern for a normal sampling distribution is

$$\text{estimate} \pm 2 \times \text{SE}.$$

Next: turn this logic into formal confidence intervals.

Four traps to avoid

1. **Confusing SD with SE.** One describes observations; the other describes estimates.
2. **Confusing the population distribution with the sampling distribution.**
3. **Thinking larger n changes the population mean.** It changes precision, not the target.
4. **Treating $n = 30$ as a law.** Normal approximation depends on shape and outliers too.

Today in four equations

Point estimate: $\bar{x}$

Sampling-distribution center: $E(\bar{X}) = \mu$

Standard error: $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

CLT: $\bar{X} \approx N\left(\text{mean} = \mu, \text{SD} = \frac{\sigma}{\sqrt{n}}\right)$

Samples vary.

Their variation is predictable.

That predictability is what makes statistical inference possible.